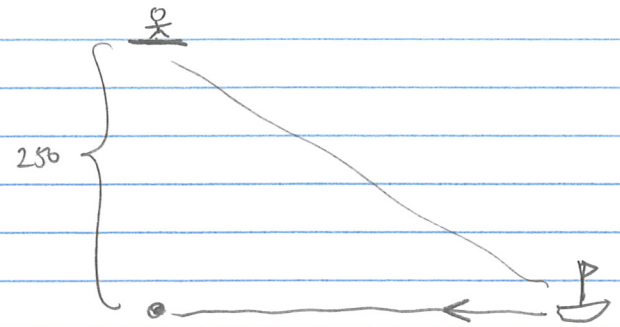
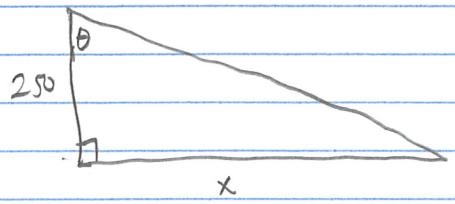
(5)





$$\tan\theta = \frac{x}{250}$$

$\frac{d}{dt}$

$$\sec^2\theta \cdot \frac{d\theta}{dt} = \frac{1}{250} \cdot \frac{dx}{dt}$$

$\frac{dx}{dt} = -20$ ← Why negative?

$x = 240$, so $\theta = \tan^{-1}\left(\frac{240}{250}\right)$

$$\sec^2\left(\tan^{-1}\left(\frac{240}{250}\right)\right) \cdot \frac{d\theta}{dt} = \frac{1}{250} \cdot (-20)$$

$$\frac{d\theta}{dt} = \frac{-20}{250\sec^2\left(\tan^{-1}\left(\frac{240}{250}\right)\right)} \approx -0.04163197335$$

The angle of the telescope decreases at 0.04 radians per sec.